

Introduction

Goal

- Measuring the dampening with various methods

Foundations

Protocoll

Sketch

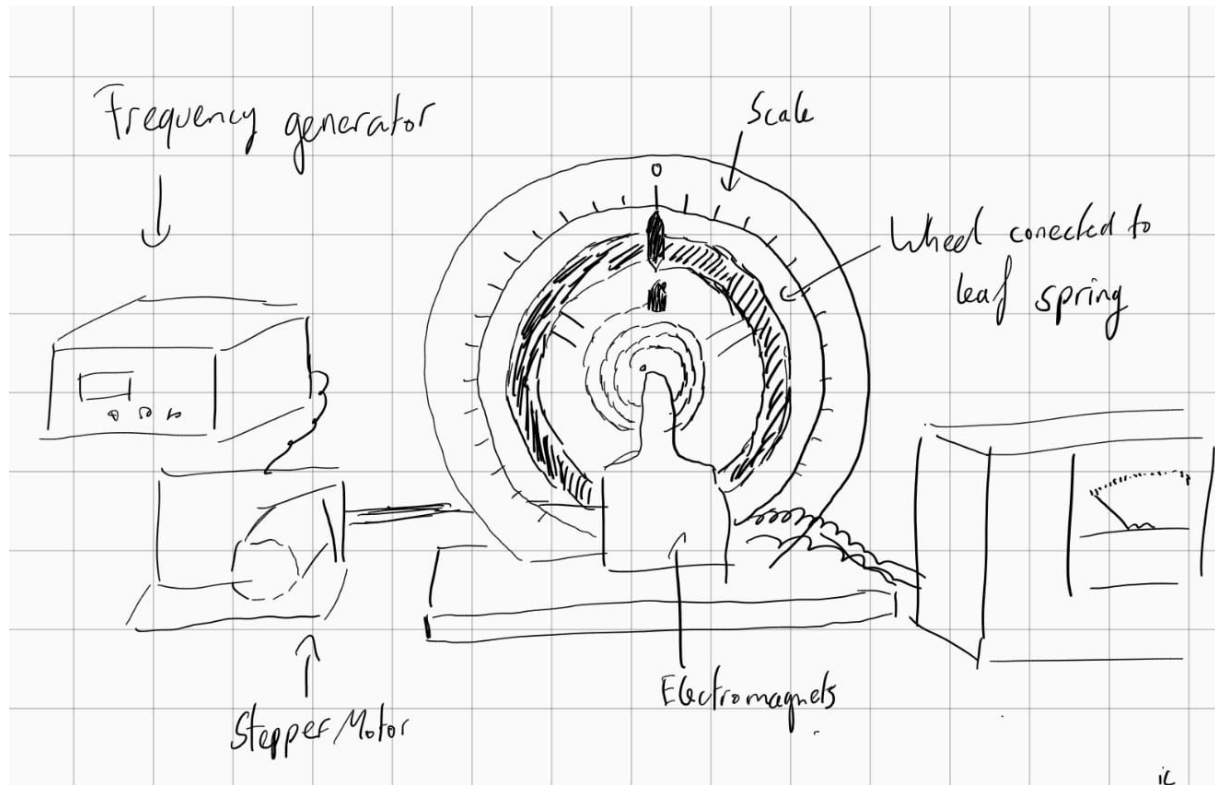


Figure 1: Sketch of the Setup

Evaluation

Determining the oscillation period

We measured 3 times 20 oscillations using a stopwatch. We assume the error due to the reaction time to be about 200 ms (-> Stopwatch error is much smaller -> negligible).

As the first step, we divided all measured times by 20 and calculated the average error of a measurement S_E .

We added the error of the mean value $S_M = \frac{S_E}{\sqrt{3}}$ to the reaction time according to the laws of error propagation:

$$\Delta T = \sqrt{S_M^2 + \left(\frac{200 \text{ ms}}{20}\right)^2} \approx 0.010 \quad (1)$$

This results in a Period of

$$T = (2.319 \pm 0.010) \text{ s} \quad (2)$$

In practice, we used the python package `uncertainties` to keep track of the errors, but I am going to write out the important steps here as well.

Determining the dampening constant with the amplitude decrease of the free oscillation

We assumed our error to be 1° . Figure 2 shows the amplitude as function of the number of oscillations. To determine the dampening constant, we performed the following steps:

1. matching regression lines through the values
2. determining the half-life of the amplitude $t_{\frac{1}{2}}$

We know from the introduction (TODO reference), that we can compute δ using (TODO Reference)

$$\delta = \frac{\ln(2)}{t_{\frac{1}{2}}} \quad (3)$$

To estimate the error $\Delta\delta$, I estimated the orange lines to be a lower and upper bound on the 1σ -Intervall of the slope of the regression lines. I determined the half-life of these error-lines as well, which gave me a upper and lower bound on the 1σ -Intervall of $t_{\frac{1}{2}}$. I then used the bound which was the farthest from the value of $t_{\frac{1}{2}}$ as an upper bound on the error of $t_{\frac{1}{2}}$, which gave me the following two values for $t_{\frac{1}{2}}$

$$40\text{mA Dampening: } t_{\frac{1}{2}} = (3.4 \pm 0.5) \text{ s} \quad (4)$$

$$55\text{mA Dampening: } t_{\frac{1}{2}} = (1.95 \pm 0.2) \text{ s} \quad (5)$$

Using Equation 3, we get

$$40\text{mA Dampening: } \delta = (0.094 \pm 0.014) \text{ s}^{-1} \quad (6)$$

$$55\text{mA Dampening: } \delta = (0.163 \pm 0.017) \text{ s}^{-1} \quad (7)$$

Error Propagation by Hand: TODO

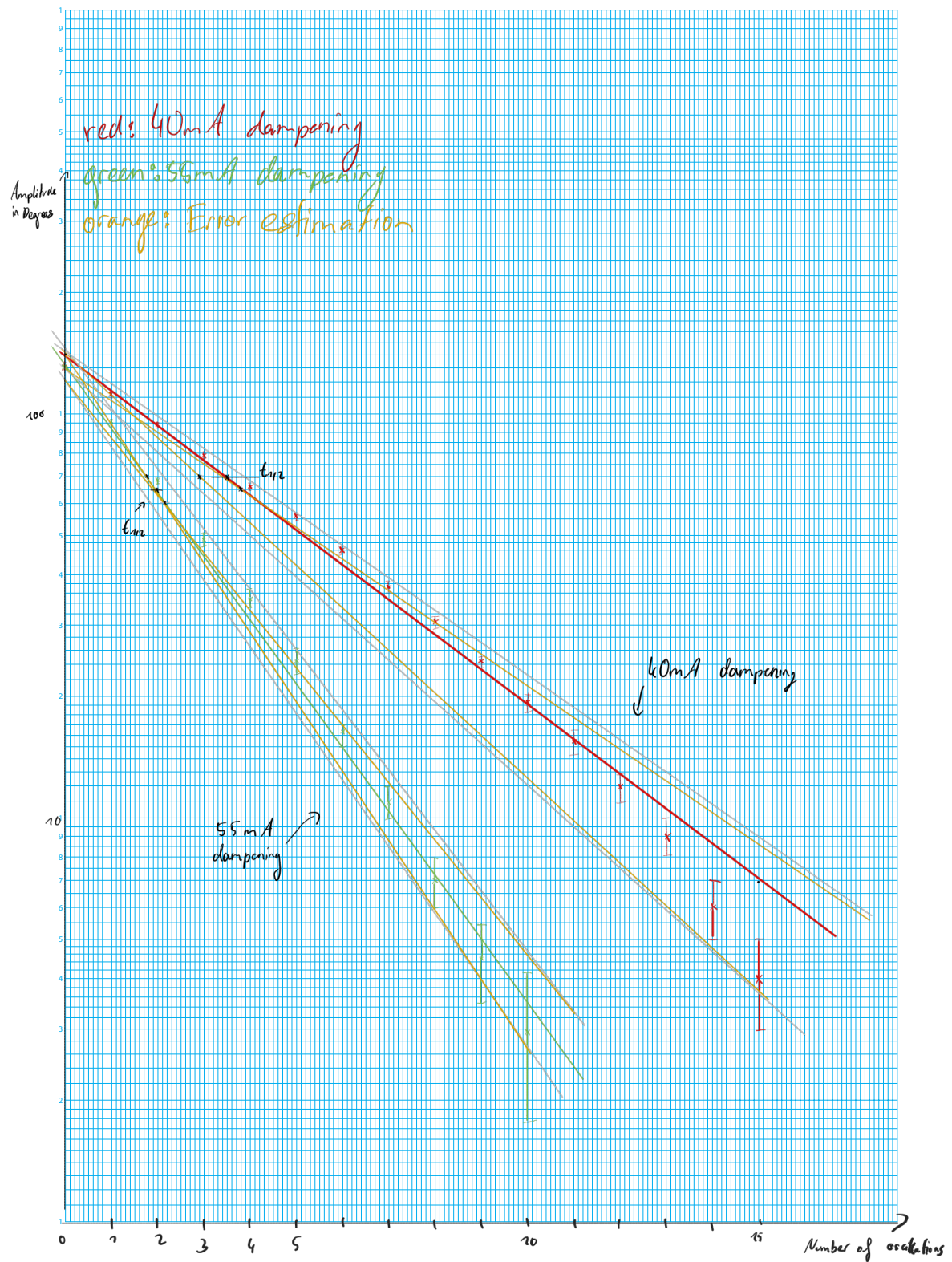


Figure 2: Amplitude as a function of the number of oscillations in a logarithmic plot

Determining the damping constant with a forced oscillation

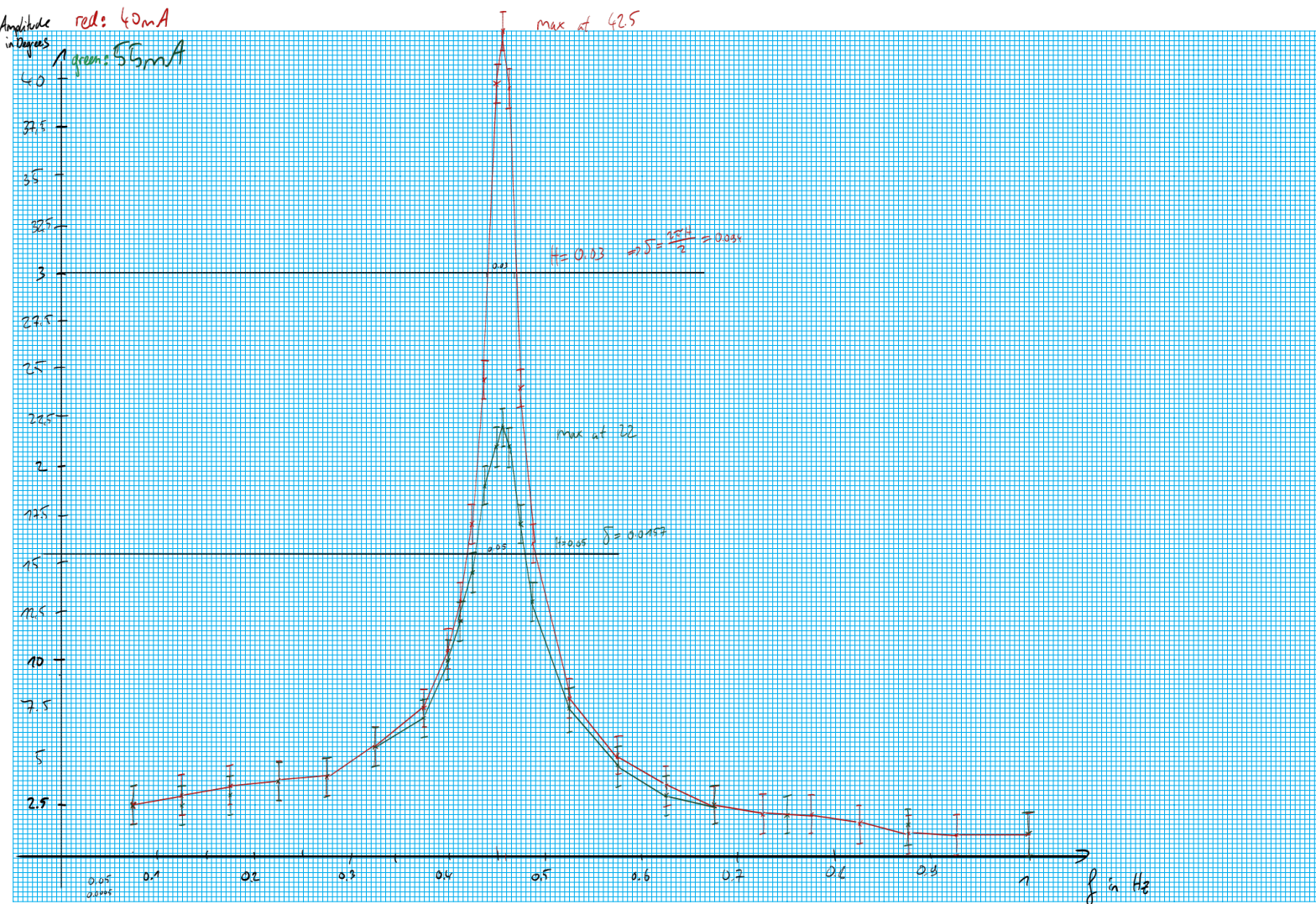


Figure 3: Amplitude as a function of the frequency of the motor

Calculation using resonance amplification

The resonance amplification is given by the ratio

$$\frac{b_{\max}}{b_0} = \frac{\omega_0}{2\delta} \quad (8)$$

with $b_{\max}$ being the maximum amplitude and b_0 being the amplitude as the frequency tends to zero.

Using our measurements for $b_{\max}$, b_0 and ω_0 , we can calculate δ this way:

	$b_{\max}$ in °	b_0 in °	δ in s^{-1}
40mA Dampening	(42.5 ± 1.0)	(2.5 ± 1.0)	(0.085 ± 0.034)
50mA Dampening	(22.0 ± 1.0)	(2.5 ± 1.0)	(0.16 ± 0.07)

Discussion

Task 2 ??

Task 4

- Error in the last new values (not very linear)
- Possible causes: Static friction at very slow speeds? (nonlinear friction)

Task 5

- Error analysis difficult (especially on paper because we can't fit different curves) (Error of frequency generator vlt)
- Passt prima mit Fehler